

Read-coherence

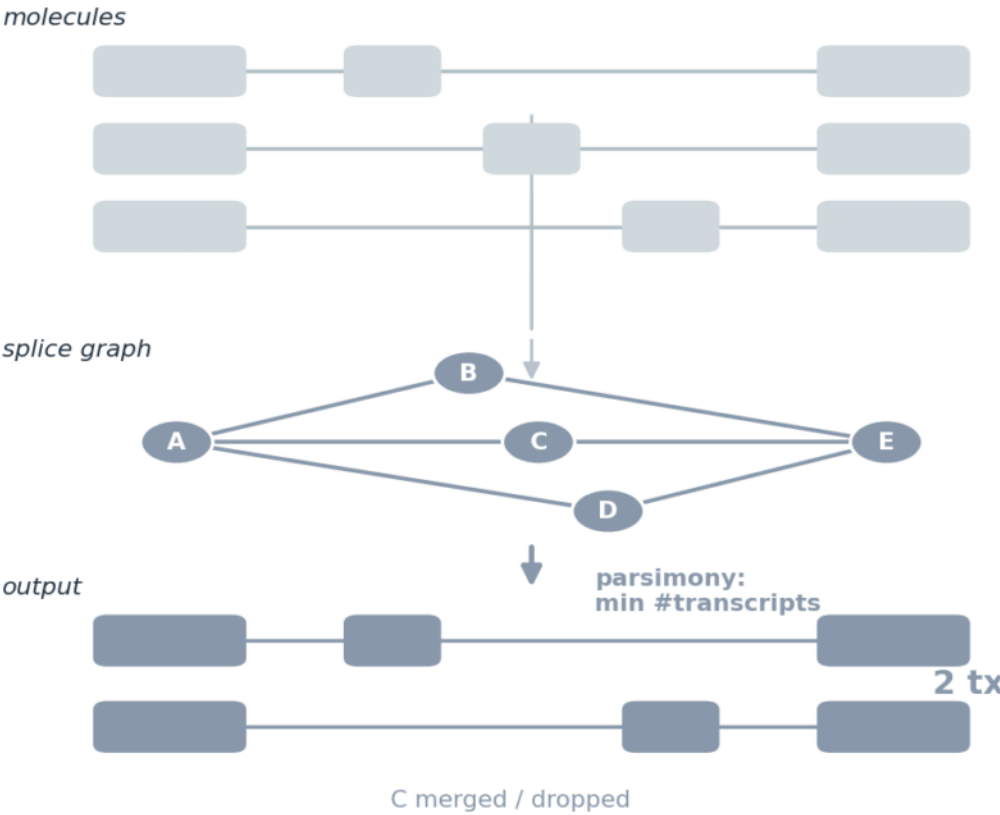
Per-molecule isoform recovery, additive over flow assembly

Recovers real isoforms StringTie's parsimony merges away —
as a provable superset of the flow output

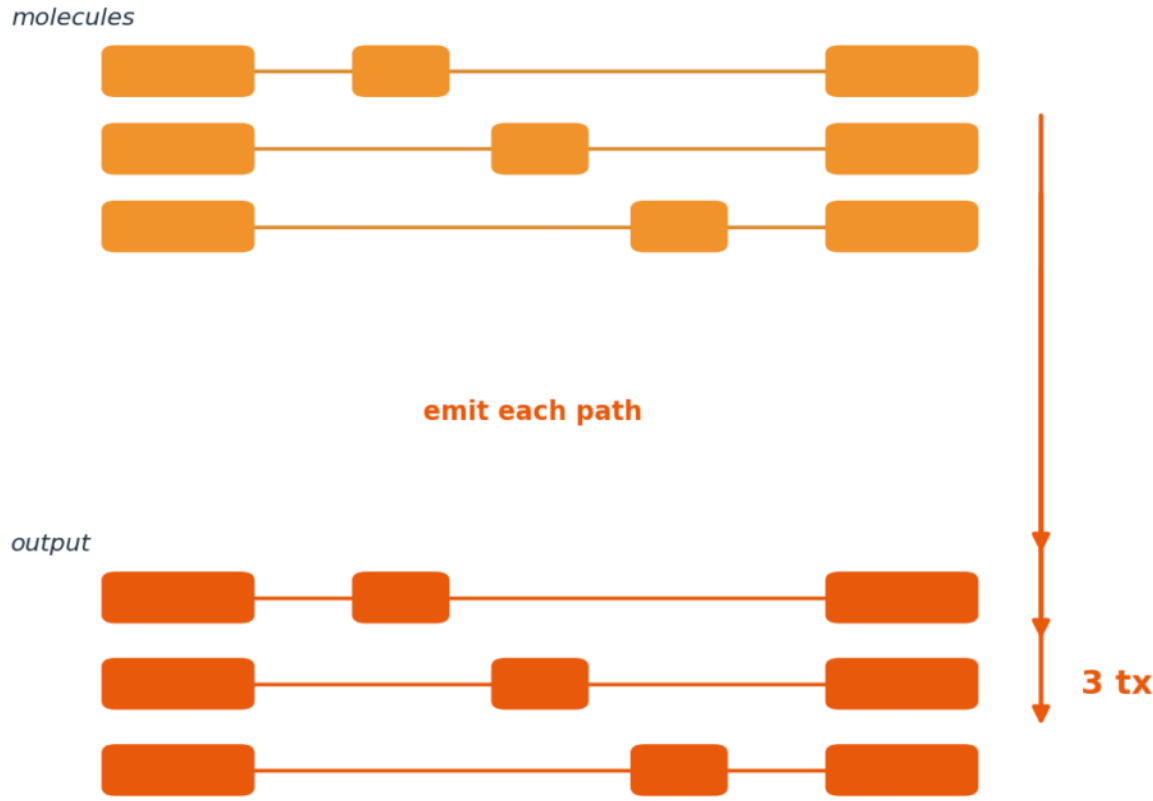
rustle · long-read transcript assembly

Why the two regimes differ

flow assembly (StringTie)

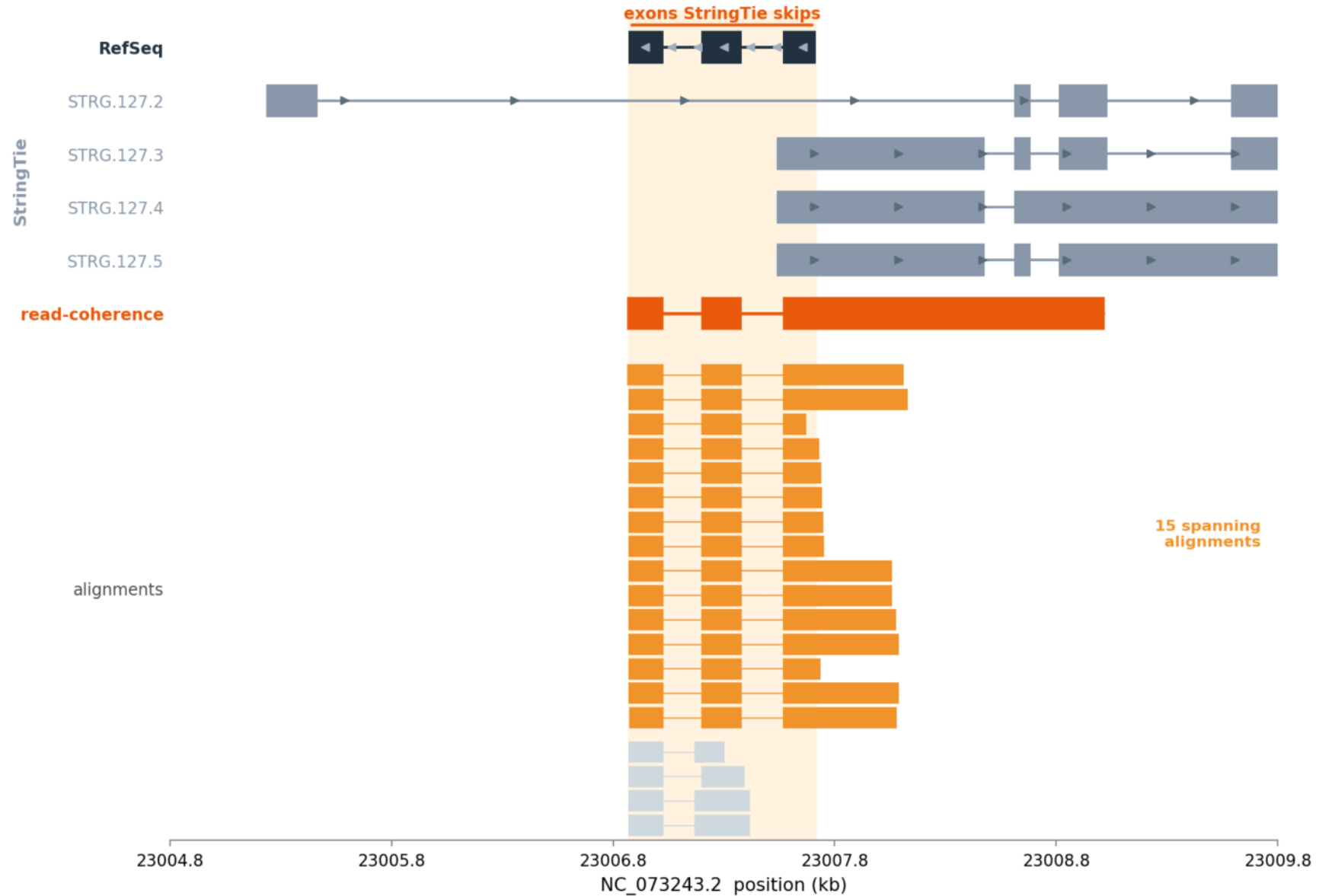


read-coherent extraction



□ molecule (alignments) ■ flow / StringTie ■ read-coherence

Read-coherence recovers a read-supported isoform StringTie skips



It's not clustering — and where transcripts begin & end

Not clustering

- Identity = exact intron chain (ordered splice junctions)
- Same junctions → same transcript; different → different
- No similarity metric, no threshold, no merge radius
- Same identity as gffcompare FSM; only knob is a read-depth floor (evidence, not a radius)

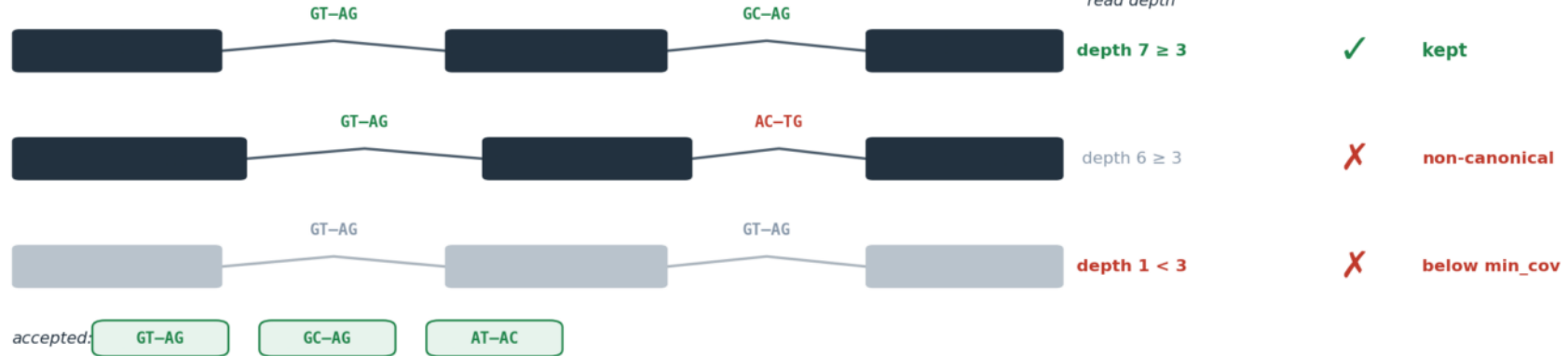
Transcript boundaries

- Junctions are base-precise (confirmed GT-AG / GC-AG / AT-AC)
- Ends = outer envelope of the supporting alignments
- Truncated fragments fold into the full-length parent
- Single-exon reads excluded (no chain → would need span-clustering)

Two precision mechanisms

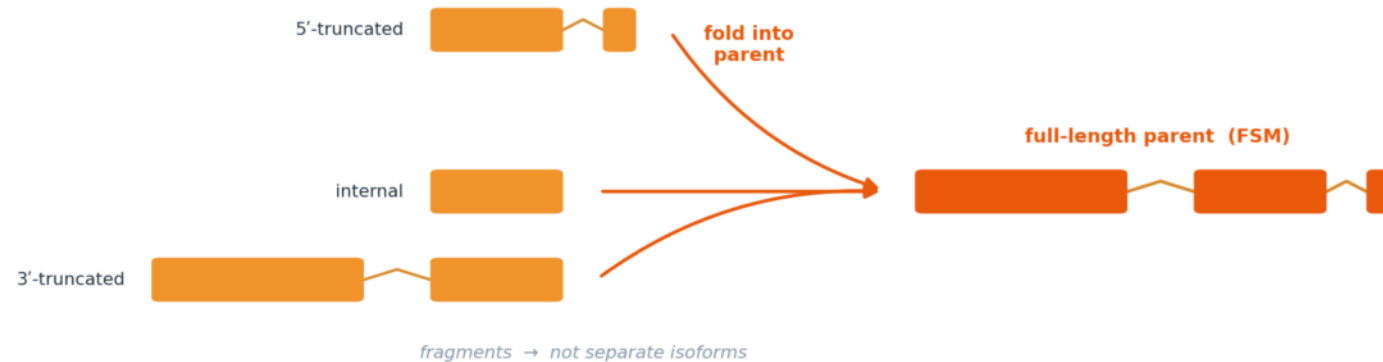
realness gate

annotation-free: genome sequence + read depth only



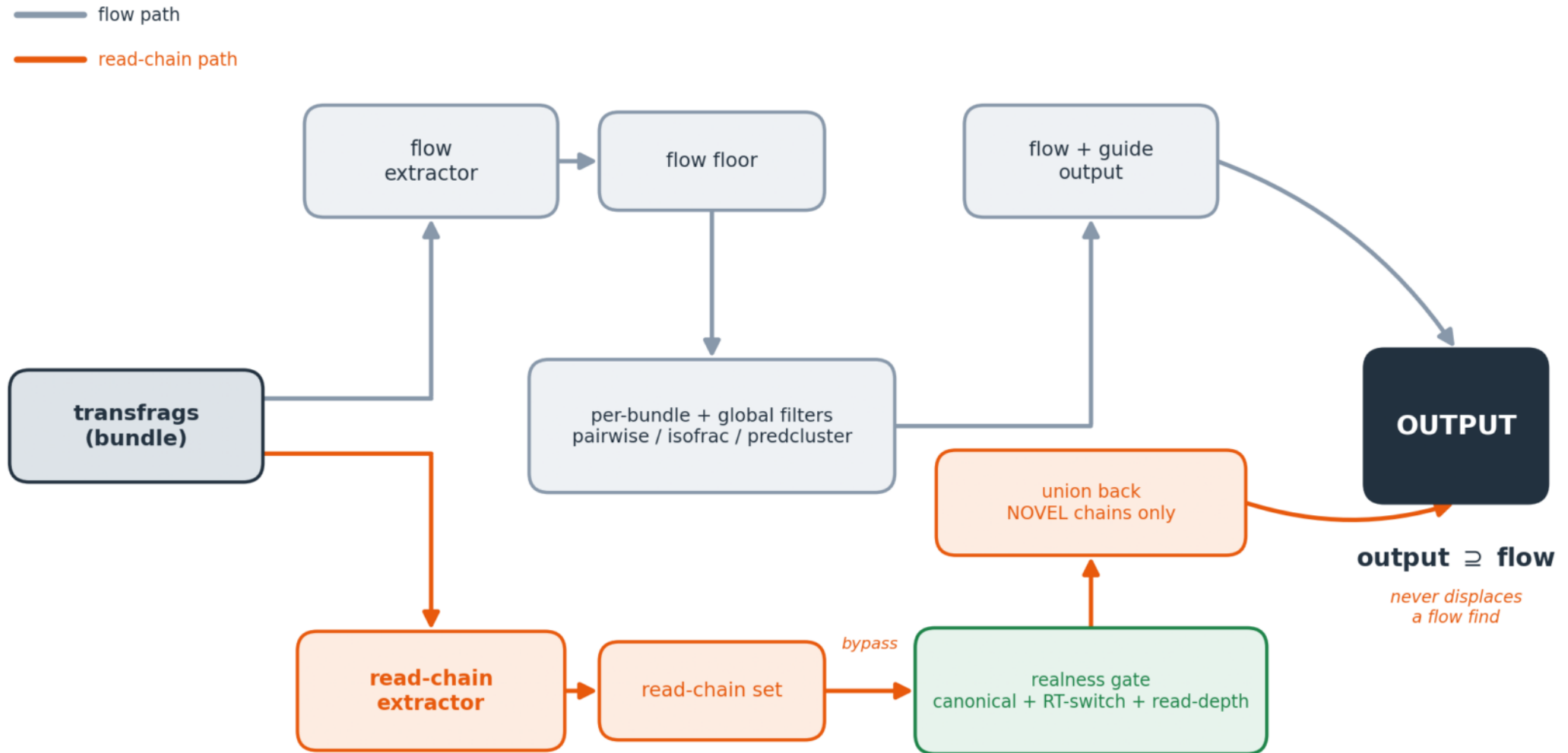
degradation-aware collapse

fragments fold into their full-length parent (ISM → FSM)

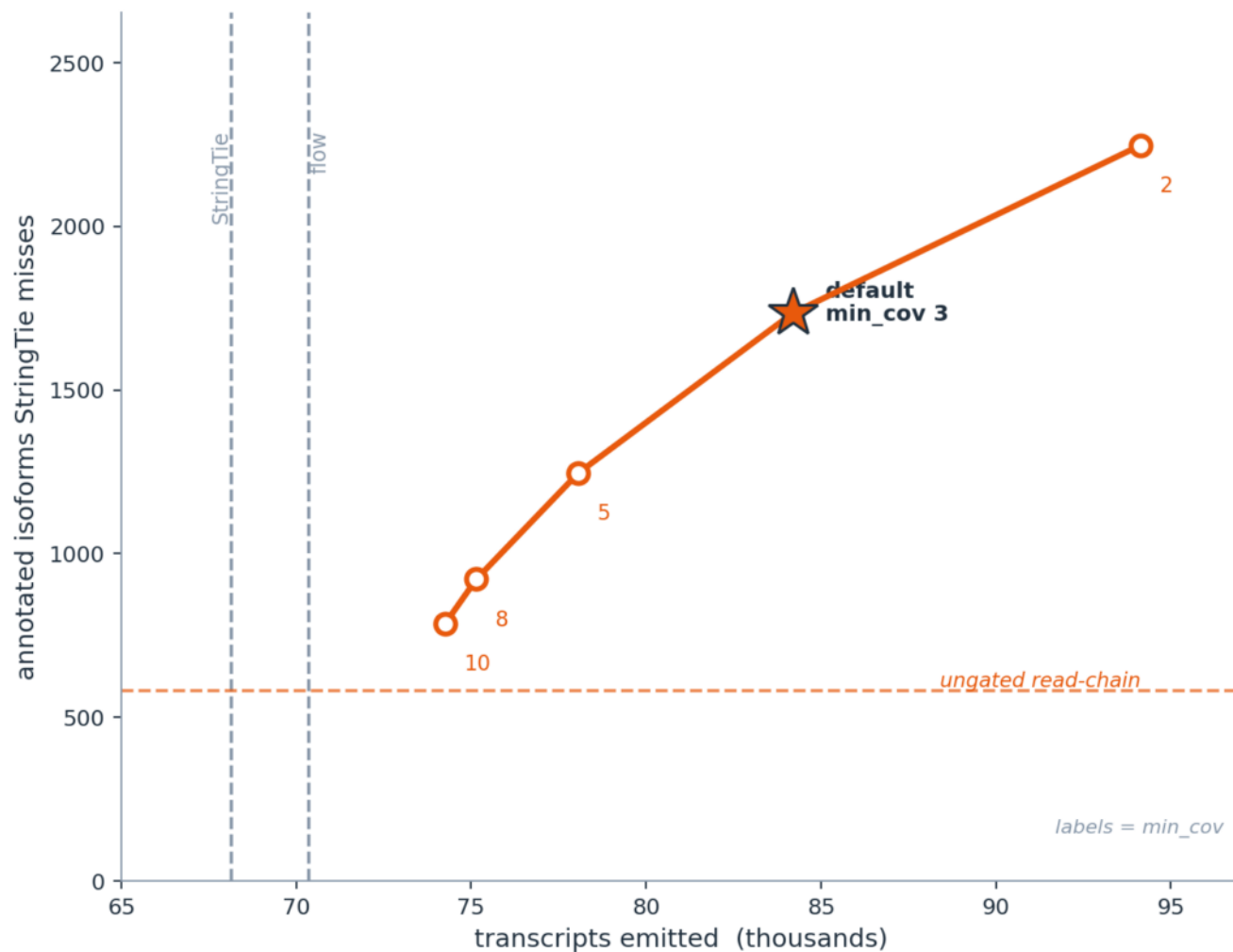


■ candidate / parent exon ■ fragment exon ■ read-coherence ● canonical / pass ● non-canonical / drop

Additive architecture: read-chain only adds



Recall vs transcript count



What the extras are (SQANTI3, n=23,791)

